

Group 6 — Data Analytics Project Report

From Raw Data to Interactive Dashboard

1. Dataset Chosen

For this project, our team selected the **Retail Sales Dataset** from Kaggle

<https://www.kaggle.com/datasets/mohammadtalib786/retail-sales-dataset>, containing **1,000 retail transactions**. The dataset includes:

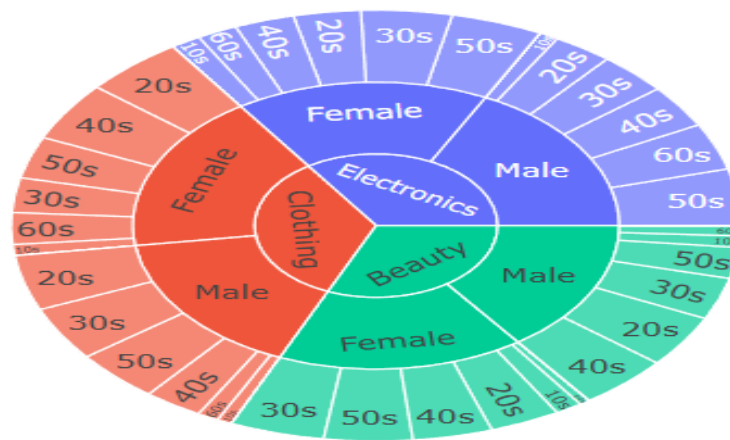
- Date of purchase
- Customer demographics (Gender, Age)
- Product Category (Electronics, Clothing, Beauty)
- Quantity purchased
- Price per Unit
- Total Amount spent

This dataset was chosen because it provides rich information for analyzing customer behavior, product performance, and sales trends all essential components of a real-world data analytics workflow.

2. Key Insights

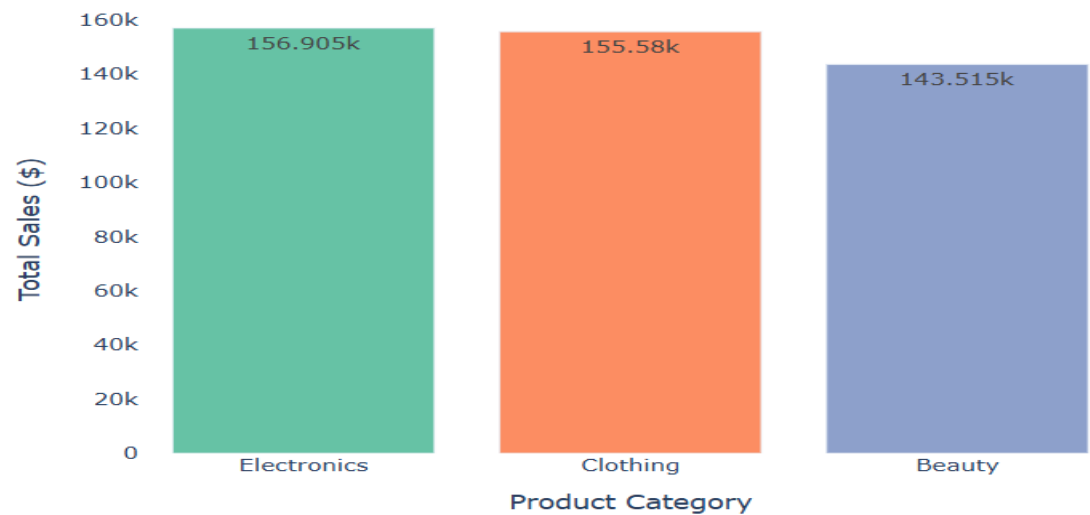
Customer Demographics (Sunburst Chart)

The sunburst chart provides a hierarchical visualization of retail sales by product category, gender, and age group. It shows how total sales are distributed across Electronics, Beauty, and Clothing, and further breaks down each category by customer gender and age group. This makes it easy to identify the customer segments contributing the most revenue and reveals purchasing patterns across demographics.



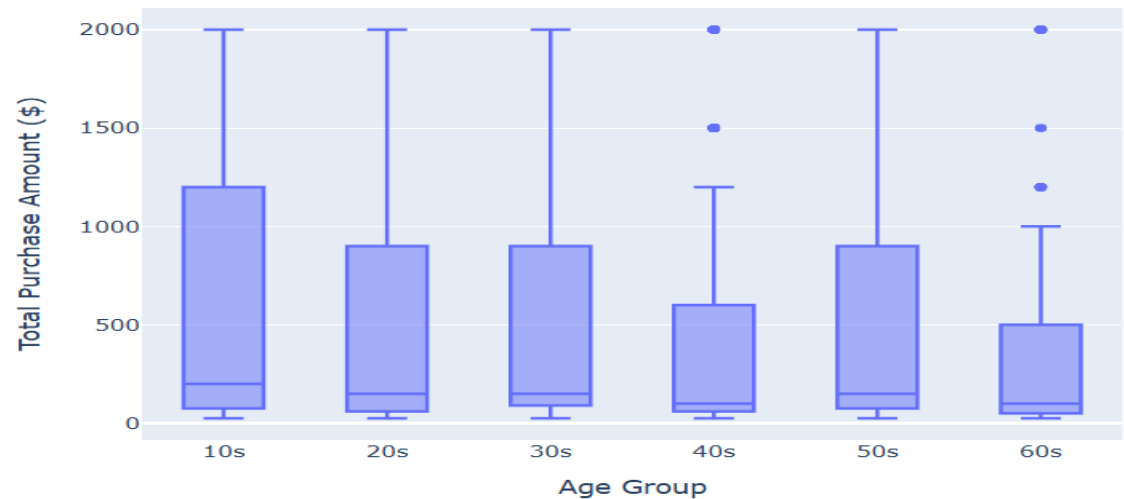
Product Category Performance -Bar Chart

The bar chart presents a clear comparison of total retail sales across Electronics, Beauty, and Clothing. By aggregating all transactions within each category and sorting them from highest to lowest, the visualization highlights which product categories generate the most revenue. This insight supports decisions related to inventory planning, marketing focus, and product strategy.



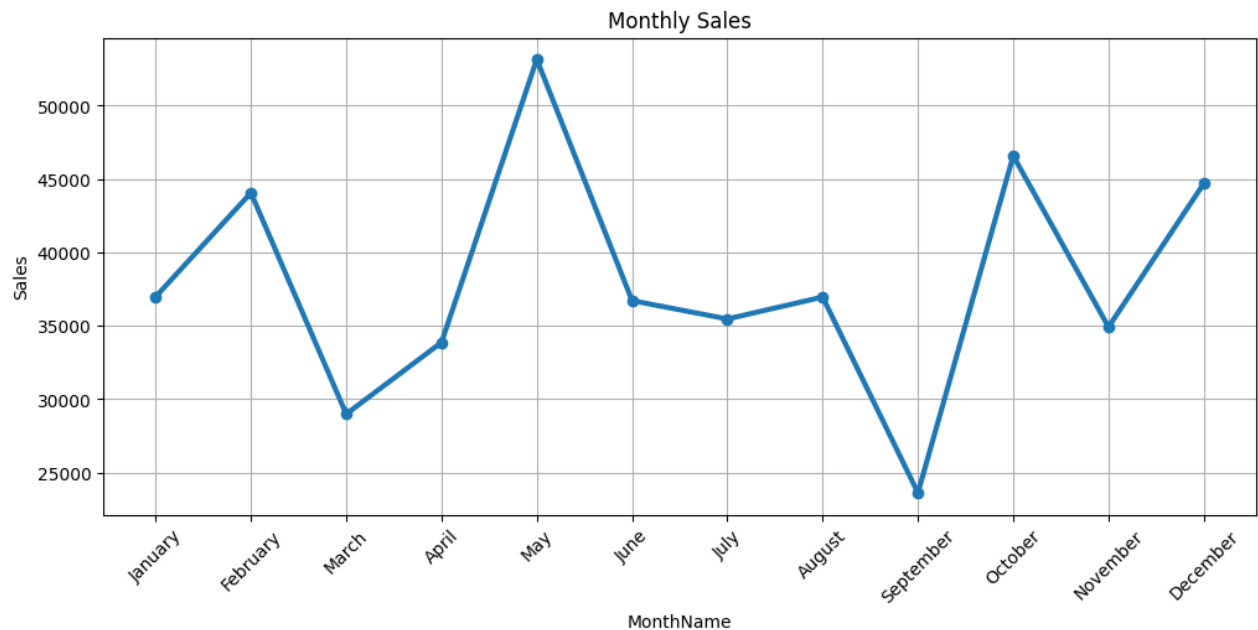
Spending Distribution by Age Group - Box Plot

The box plot compares spending behavior across age groups using quartiles, medians, whiskers, and outliers. It highlights typical spending levels, variation within each group, and extreme purchase amounts. This helps identify which age groups spend more, which show stable behaviour, and which have large variations.



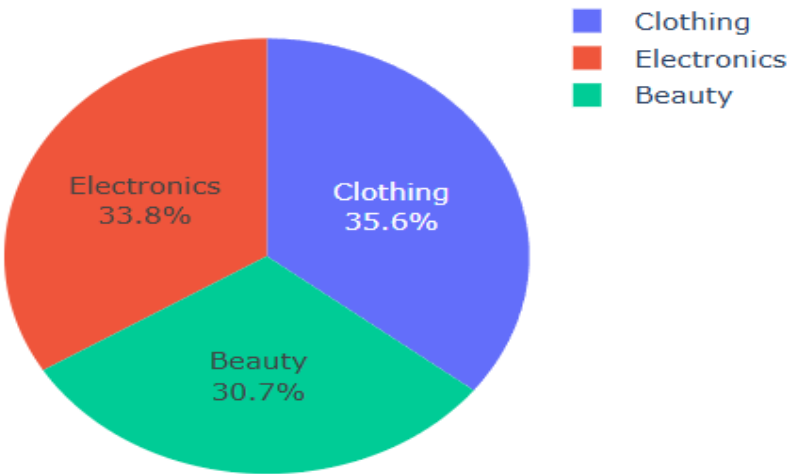
Monthly Sales Trends -Line Chart

The line chart shows how total sales fluctuate month-to-month. It reveals peak spending months, seasonal patterns, and periods of low activity — insights valuable for forecasting and planning promotions.



Quantity Purchased per Category - Pie Chart

The pie chart summarizes total units purchased across product categories. It highlights which categories have the highest purchase volume and reveals customer preference for product types.



3. Challenges Faced

During the project, we encountered several challenges:

- **Dashboard Integration:** Combining multiple charts into a single Dash callback and ensuring they updated dynamically based on filters.
- **Layout & Styling:** Designing a clean, responsive dashboard layout with KPI cards and a 3×2 grid of charts.
- **Ngrok Deployment:** Setting up ngrok in Google Colab to render the dashboard externally required troubleshooting authentication and port configuration.

4. What We Learned

This project helped us gain hands-on experience with:

- Real-world **data cleaning and wrangling**
- **Exploratory Data Analysis (EDA)** using pandas, matplotlib, seaborn, and Plotly
- **Interactive dashboard development** using Dash
- Building **dynamic visualizations** that respond to user filters
- **Team collaboration** in dividing tasks and integrating multiple components
- Presenting insights clearly through charts and narrative explanations

Overall, we learned how to transform raw data into meaningful insights and present them through a professional, interactive dashboard — a skill directly applicable to real data analyst roles.